IMPORTS

```
import os
import numpy as np
import tensorflow as tf

import matplotlib.pyplot as plt
import seaborn as sns

from google.colab import drive

from tensorflow.keras.applications import ConvNeXtTiny
from tensorflow.keras.applications.convnext import preprocess_input

from sklearn.utils.class_weight import compute_class_weight
from sklearn.metrics import confusion_matrix, classification_report
```

MOUNT DRIVE

```
drive.mount("/content/drive")
```

```
Mounted at /content/drive
```

PATHS (EDIT)

```
TRAIN_DIR = "/content/drive/MyDrive/Project work/Dataset/Epic and CSCR hospital Dataset_clean/Train"
TEST_DIR  = "/content/drive/MyDrive/Project work/Dataset/Epic and CSCR hospital Dataset_clean/Test"

MODEL_PATH = "/content/drive/MyDrive/Models/convnext_tiny_tumor.keras"
```

```
os.path.exists(TRAIN_DIR)
os.path.exists(TEST_DIR)
```

```
True
```

CONFIG

```
IMG_SIZE = 384
BATCH_SIZE = 16
SEED = 42

WARMUP_EPOCHS = 3
HEAD_EPOCHS   = 5
FT1_EPOCHS    = 6
FT2_EPOCHS    = 4

TARGET_ACC = 0.90

GRAPHS=[]
```

TRAIN + VAL

```
train_raw = tf.keras.utils.image_dataset_from_directory(
    TRAIN_DIR,
    validation_split=0.2,
    subset="training",
    seed=SEED,
    image_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
```

```
    label_mode="categorical"
)

val_raw = tf.keras.utils.image_dataset_from_directory(
    TRAIN_DIR,
    validation_split=0.2,
    subset="validation",
    seed=SEED,
    image_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    label_mode="categorical"
)

class_names = train_raw.class_names
NUM_CLASSES = len(class_names)

print("Classes:", class_names)
```

```
Found 9650 files belonging to 4 classes.
Using 7720 files for training.
Found 9650 files belonging to 4 classes.
Using 1930 files for validation.
Classes: ['glioma', 'meningioma', 'notumor', 'pituitary']
```

TEST

```
test_raw = tf.keras.utils.image_dataset_from_directory(
    TEST_DIR,
    image_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    label_mode="categorical"
)
```

```
Found 2180 files belonging to 4 classes.
```

AUGMENT

```
augment = tf.keras.Sequential([
    tf.keras.layers.RandomFlip("horizontal"),
    tf.keras.layers.RandomRotation(0.1),
    tf.keras.layers.RandomZoom(0.15),
    tf.keras.layers.RandomContrast(0.15)
])
```

PREPROCESS

```
AUTOTUNE = tf.data.AUTOTUNE


def preprocess(x, y):
    x = preprocess_input(x)
    return x, y


def augment_and_preprocess(x, y):
    x = augment(x, training=True)
    x = preprocess_input(x)
    return x, y
```

FINAL PIPELINES

```
train_ds = train_raw.map(
    augment_and_preprocess,
    num_parallel_calls=AUTOTUNE
)

val_ds = val_raw.map(
    preprocess,
    num_parallel_calls=AUTOTUNE
)

test_ds = test_raw.map(
    preprocess,
    num_parallel_calls=AUTOTUNE
)


train_ds = train_ds.prefetch(AUTOTUNE)
val_ds   = val_ds.prefetch(AUTOTUNE)
test_ds  = test_ds.prefetch(AUTOTUNE)
```

CLASS WEIGHTS

```
labels = []

for _, y in train_raw:
    labels.extend(np.argmax(y.numpy(), axis=1))

labels = np.array(labels)

weights = compute_class_weight(
    class_weight="balanced",
    classes=np.unique(labels),
    y=labels
)

class_weights = dict(enumerate(weights))

print("Class weights:", class_weights)
```

```
Class weights: {0: np.float64(0.8004977187888843), 1: np.float64(1.107921928817451), 2: np.float64(1.238767650834403), 3: np.flo
```

CALLBACKS

```
callbacks = [

    tf.keras.callbacks.ModelCheckpoint(
        "best_stage.h5",
        monitor="val_loss",
        save_best_only=True
    ),

    tf.keras.callbacks.EarlyStopping(
        monitor="val_loss",
        patience=3,
        restore_best_weights=True
    ),

    tf.keras.callbacks.ReduceLROnPlateau(
        monitor="val_loss",
        patience=2,
        factor=0.3,
        min_lr=1e-6
    )
]
```

BUILD

```
base_model = ConvNeXtTiny(
    include_top=False,
    weights="imagenet",
    input_shape=(IMG_SIZE, IMG_SIZE, 3)
)

base_model.trainable = False


inputs = tf.keras.Input((IMG_SIZE, IMG_SIZE, 3))

x = base_model(inputs, training=False)

x = tf.keras.layers.GlobalAveragePooling2D()(x)
x = tf.keras.layers.BatchNormalization()(x)

x = tf.keras.layers.Dense(256, activation="gelu")(x)
x = tf.keras.layers.Dropout(0.3)(x)

outputs = tf.keras.layers.Dense(
    NUM_CLASSES,
    activation="softmax"
)(x)


model = tf.keras.Model(inputs, outputs)

model.summary()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/convnext/convnext_tiny_notop.h5
111650432/111650432 ━━━━━━━━━━━━━━━━━━━━ 6s 0us/step
Model: "functional_5"
```

Layer (type)	Output Shape	Param #
input_layer_6 (InputLayer)	(None, 384, 384, 3)	0
convnext_tiny (Functional)	(None, 12, 12, 768)	27,820,128
global_average_pooling2d (GlobalAveragePooling2D)	(None, 768)	0
batch_normalization (BatchNormalization)	(None, 768)	3,072
dense (Dense)	(None, 256)	196,864
dropout (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 4)	1,028

```
Total params: 28,021,092 (106.89 MB)
Trainable params: 199,428 (779.02 KB)
Non-trainable params: 27,821,664 (106.13 MB)
```

LOSS

```
loss_fn = tf.keras.losses.CategoricalCrossentropy(label_smoothing=0.1)

metrics = [
    "accuracy",
    tf.keras.metrics.Precision(),
    tf.keras.metrics.Recall()
]
```

TRAINING

WARMUP

```
print("Warmup...")
```

```
model.compile(
    optimizer=tf.keras.optimizers.AdamW(1e-3),
    loss=loss_fn,
    metrics=metrics
)

hist_warmup = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=WARMUP_EPOCHS,
    callbacks=callbacks,
    class_weight=class_weights
)

model.load_weights("best_stage.h5")
GRAPHS.append(hist_warmup)
```

```
Warmup...
Epoch 1/3
483/483 ━━━━━━━━━━━━━━━━━━━━ 0s 115ms/step - accuracy: 0.7369 - loss: 0.9216 - precision: 0.7742 - recall: 0.6930WARNING:absl:Yo
483/483 ━━━━━━━━━━━━━━━━━━━━ 178s 307ms/step - accuracy: 0.7370 - loss: 0.9214 - precision: 0.7743 - recall: 0.6931 - val_accura
Epoch 2/3
482/483 ━━━━━━━━━━━━━━━━━━━━ 0s 83ms/step - accuracy: 0.8701 - loss: 0.6336 - precision: 0.8980 - recall: 0.8265WARNING:absl:You
483/483 ━━━━━━━━━━━━━━━━━━━━ 50s 101ms/step - accuracy: 0.8701 - loss: 0.6335 - precision: 0.8981 - recall: 0.8266 - val_accurac
Epoch 3/3
482/483 ━━━━━━━━━━━━━━━━━━━━ 0s 84ms/step - accuracy: 0.8971 - loss: 0.5874 - precision: 0.9238 - recall: 0.8614WARNING:absl:You
483/483 ━━━━━━━━━━━━━━━━━━━━ 50s 102ms/step - accuracy: 0.8971 - loss: 0.5873 - precision: 0.9238 - recall: 0.8614 - val_accurac
```

HEAD

```
print("Head Training...")

hist_head = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=HEAD_EPOCHS,
    callbacks=callbacks,
    class_weight=class_weights
)

model.load_weights("best_stage.h5")
GRAPHS.append(hist_head)
```

```
Head Training...
Epoch 1/5
482/483 ━━━━━━━━━━━━━━━━━━━━ 0s 84ms/step - accuracy: 0.9149 - loss: 0.5522 - precision: 0.9394 - recall: 0.8845WARNING:absl:You
483/483 ━━━━━━━━━━━━━━━━━━━━ 51s 102ms/step - accuracy: 0.9149 - loss: 0.5522 - precision: 0.9394 - recall: 0.8845 - val_accurac
Epoch 2/5
482/483 ━━━━━━━━━━━━━━━━━━━━ 0s 84ms/step - accuracy: 0.9191 - loss: 0.5425 - precision: 0.9423 - recall: 0.8845WARNING:absl:You
483/483 ━━━━━━━━━━━━━━━━━━━━ 50s 102ms/step - accuracy: 0.9192 - loss: 0.5425 - precision: 0.9423 - recall: 0.8846 - val_accurac
Epoch 3/5
482/483 ━━━━━━━━━━━━━━━━━━━━ 0s 83ms/step - accuracy: 0.9317 - loss: 0.5332 - precision: 0.9490 - recall: 0.9023WARNING:absl:You
483/483 ━━━━━━━━━━━━━━━━━━━━ 50s 101ms/step - accuracy: 0.9317 - loss: 0.5332 - precision: 0.9490 - recall: 0.9024 - val_accurac
Epoch 4/5
482/483 ━━━━━━━━━━━━━━━━━━━━ 0s 83ms/step - accuracy: 0.9282 - loss: 0.5197 - precision: 0.9502 - recall: 0.9026WARNING:absl:You
483/483 ━━━━━━━━━━━━━━━━━━━━ 50s 101ms/step - accuracy: 0.9282 - loss: 0.5197 - precision: 0.9502 - recall: 0.9026 - val_accurac
Epoch 5/5
483/483 ━━━━━━━━━━━━━━━━━━━━ 49s 100ms/step - accuracy: 0.9400 - loss: 0.5086 - precision: 0.9578 - recall: 0.9103 - val_accurac
```

FT1 (50%)

```
print("Fine-tune 50%...")


n = len(base_model.layers)
unfreeze = int(0.5 * n)

for layer in base_model.layers[unfreeze:]:
    layer.trainable = True


model.compile(
```

```
    optimizer=tf.keras.optimizers.AdamW(5e-5),
    loss=loss_fn,
    metrics=metrics
)


hist_ft1 = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=FT1_EPOCHS,
    callbacks=callbacks,
    class_weight=class_weights
)

model.load_weights("best_stage.h5")
GRAPHS.append(hist_ft1)
```

```
Fine-tune 50%...
Epoch 1/6
483/483 ━━━━━━━━━━━━━━━━━━━━ 0s 332ms/step - accuracy: 0.9500 - loss: 0.4974 - precision: 0.9580 - recall: 0.9168WARNING:absl:Yo
483/483 ━━━━━━━━━━━━━━━━━━━━ 220s 372ms/step - accuracy: 0.9500 - loss: 0.4974 - precision: 0.9581 - recall: 0.9168 - val_accura
Epoch 2/6
483/483 ━━━━━━━━━━━━━━━━━━━━ 0s 290ms/step - accuracy: 0.9779 - loss: 0.4420 - precision: 0.9858 - recall: 0.9678WARNING:absl:Yo
483/483 ━━━━━━━━━━━━━━━━━━━━ 151s 309ms/step - accuracy: 0.9779 - loss: 0.4420 - precision: 0.9858 - recall: 0.9678 - val_accura
Epoch 3/6
483/483 ━━━━━━━━━━━━━━━━━━━━ 0s 290ms/step - accuracy: 0.9863 - loss: 0.4278 - precision: 0.9896 - recall: 0.9766WARNING:absl:Yo
483/483 ━━━━━━━━━━━━━━━━━━━━ 151s 309ms/step - accuracy: 0.9863 - loss: 0.4277 - precision: 0.9896 - recall: 0.9766 - val_accura
Epoch 4/6
483/483 ━━━━━━━━━━━━━━━━━━━━ 0s 290ms/step - accuracy: 0.9939 - loss: 0.4120 - precision: 0.9958 - recall: 0.9851WARNING:absl:Yo
483/483 ━━━━━━━━━━━━━━━━━━━━ 151s 309ms/step - accuracy: 0.9939 - loss: 0.4120 - precision: 0.9958 - recall: 0.9851 - val_accura
Epoch 5/6
483/483 ━━━━━━━━━━━━━━━━━━━━ 0s 290ms/step - accuracy: 0.9953 - loss: 0.4028 - precision: 0.9975 - recall: 0.9920WARNING:absl:Yo
483/483 ━━━━━━━━━━━━━━━━━━━━ 150s 309ms/step - accuracy: 0.9953 - loss: 0.4028 - precision: 0.9975 - recall: 0.9920 - val_accura
Epoch 6/6
483/483 ━━━━━━━━━━━━━━━━━━━━ 0s 290ms/step - accuracy: 0.9973 - loss: 0.3971 - precision: 0.9982 - recall: 0.9925WARNING:absl:Yo
483/483 ━━━━━━━━━━━━━━━━━━━━ 150s 309ms/step - accuracy: 0.9973 - loss: 0.3971 - precision: 0.9982 - recall: 0.9925 - val_accura
```

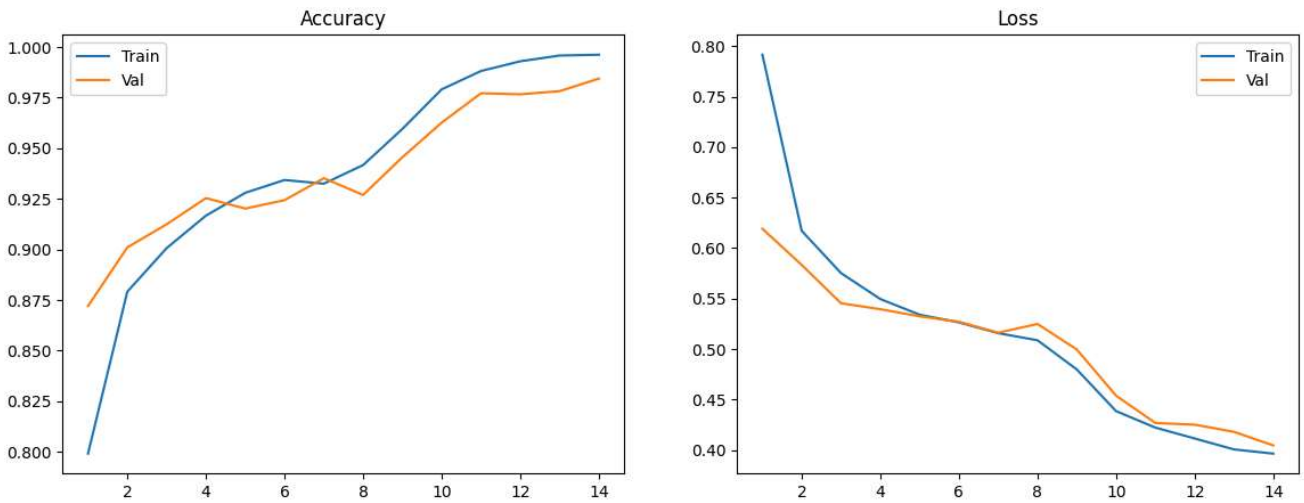
FT2 (IF < 90%)

```
best_val = max(hist_ft1.history["val_accuracy"])

print("Best Val:", best_val)


if best_val < TARGET_ACC:

    print("Full Fine-tuning...")

    base_model.trainable = True


    model.compile(
        optimizer=tf.keras.optimizers.AdamW(2e-5),
        loss=loss_fn,
        metrics=metrics
    )


    hist_ft2 = model.fit(
        train_ds,
        validation_data=val_ds,
        epochs=FT2_EPOCHS,
        callbacks=callbacks,
        class_weight=class_weights
    )
    GRAPHS.append(hist_ft2)
    model.load_weights("best_stage.h5")
```

```
Best Val: 0.984455943107605
```

GRAPH

```
def plot_history(histories):

    acc, val_acc, loss, val_loss = [], [], [], []

    for h in histories:

        acc += h.history["accuracy"]
        val_acc += h.history["val_accuracy"]

        loss += h.history["loss"]
        val_loss += h.history["val_loss"]


    e = range(1, len(acc)+1)


    plt.figure(figsize=(14,5))

    plt.subplot(1,2,1)
    plt.plot(e, acc, label="Train")
    plt.plot(e, val_acc, label="Val")
    plt.legend()
    plt.title("Accuracy")

    plt.subplot(1,2,2)
    plt.plot(e, loss, label="Train")
    plt.plot(e, val_loss, label="Val")
    plt.legend()
    plt.title("Loss")

    plt.show()

plot_history(GRAPHS)
```



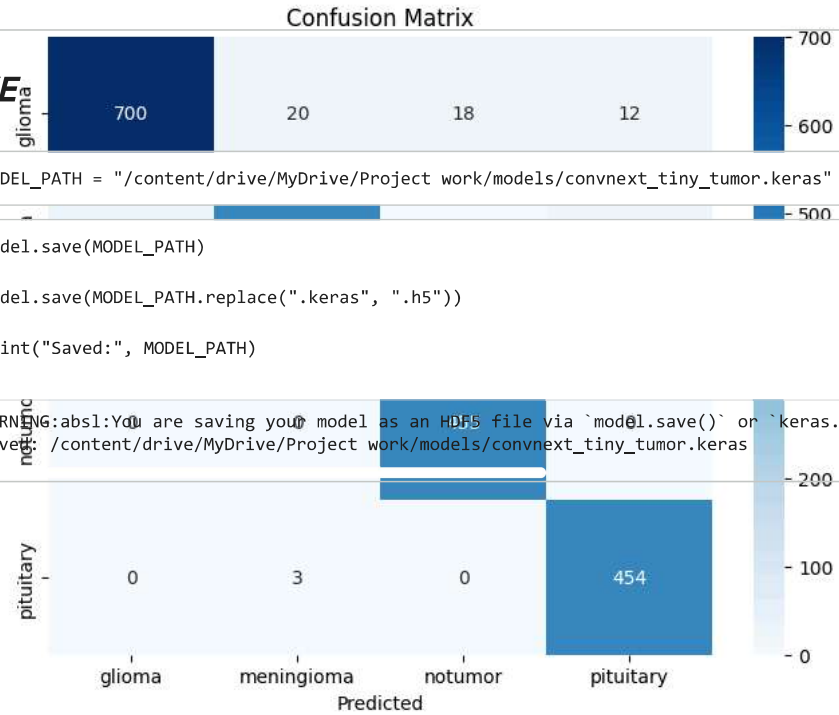
TEST

```
model.evaluate(test_ds)
```

```
137/137 ━━━━━━━━━━━━━━━━━━━━ 137s 994ms/step - accuracy: 0.9485 - loss: 0.4670 - precision: 0.9635 - recall: 0.9357
[0.4629051983356476,
 0.9504587054252625,
 0.9645892381668091,
 0.9371559619903564]
```

CONFUSION MATRIX

```
y_true = []
y_pred = []


for x, y in test_ds:

    p = model.predict(x, verbose=0)

    y_true.extend(np.argmax(y, axis=1))
    y_pred.extend(np.argmax(p, axis=1))


cm = confusion_matrix(y_true, y_pred)


plt.figure(figsize=(8,6))

sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=class_names,
    yticklabels=class_names
)

plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("True")

plt.show()


print(classification_report(
    y_true,
    y_pred,
    target_names=class_names
))
```



SAVE

```
MODEL_PATH = "/content/drive/MyDrive/Project work/models/convnext_tiny_tumor.keras"
```

```
model.save(MODEL_PATH)

model.save(MODEL_PATH.replace(".keras", ".h5"))

print("Saved:", MODEL_PATH)
```

```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format
Saved: /content/drive/MyDrive/Project work/models/convnext_tiny_tumor.keras
```

	precision	recall	f1-score	support
glioma	0.95	0.93	0.94	750
meningioma	0.95	0.89	0.92	518
notumor	0.96	1.00	0.98	455
pituitary	0.94	0.99	0.96	457
accuracy			0.95	2180
macro avg	0.95	0.96	0.95	2180
weighted avg	0.95	0.95	0.95	2180